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JavaScript Iterables

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An Iterable is an Iterable Object

An iterable can be iterated over with `for..of` loops

The For Of Loop

The JavaScript `for..of` statement loops through the elements of an iterable object.

Syntax

```
for (variable of iterable) {
  // code block to be executed
}
```

Iterating

Iterating means looping over a sequence of elements.

Here are some easy examples:

- Iterating over a String
- Iterating over an Array
- Iterating over a Set
- Iterating over a Map

Example

```
const name = "W3Schools";

for (const x of name) {
  // code block to be executed
}
```

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Iterating Over an Array

You can use a `for..of` loop to iterate over the elements of an Array:

Example 1

```
const letters = ["a","b","c"];

for (const x of letters) {
  // code block to be executed
}
```

Try it Yourself »

Example 2

```
const numbers = [2,4,6,8];

for (const x of numbers) {
  // code block to be executed
}
```

Try it Yourself »

Example

```
const letters = new Set(["a","b","c"]);

for (const x of letters) {
  // code block to be executed
}
```

Try it Yourself »

Iterating Over a Map

You can use a `for..of` loop to iterate over the elements of a Map:

Example

```
const fruits = new Map([
  ["apples", 500],
  ["bananas", 300],
  ["oranges", 200]
]);

for (const x of fruits) {
  // code block to be executed
}
```

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JavaScript Iterators

The **iterator protocol** defines how to produce a **sequence of values** from an object.

An object becomes an **iterator** when it implements a `next()` method.

The `next()` method must return an object with two properties:

- value (the next value)
- done (true or false)

value	The value returned by the iterator (Can be omitted if done is true)
done	*true* if the iterator has completed *false* if the iterator has produced a new value

Note

Technically, iterables must implement the `Symbol.iterator` method.

In JavaScript the following are iterables:

- Strings
- Arrays
- Typed Arrays
- Sets
- Maps

Because their prototype objects have a `Symbol.iterator` method:

Home Made Iterable

This iterable returns never ending number: 10,20,30,40,.... when `the next()` method is called:

Example

```
// Home Made Iterable
function myNumbers() {
```

```
      return {value:n, done:false};
    }
  };
}

// Run the Iterable
const n = myNumbers();
n.next(); // Returns 10
n.next(); // Returns 20
n.next(); // Returns 30
```

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Note

Home made iterables do not support the JavaScript `for..of` statement.

A JavaScript iterable is an object that has a **Symbol.iterator**.

The `Symbol.iterator` is a function that returns a `next()` function.

An iterable can be iterated over with the code: `for (const x of iterable) { }`

Example

```
// Create an Object
myNumbers = {};

// Make it Iterable
myNumbers[Symbol.iterator] = function() {
  let n = 0;
  done = false;
  return {
    next() {
      n += 10;
      if (n == 100) {done = true}
      return {value:n, done:done};
    }
  };
}
```

```
}
```

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The Symbol.iterator method is called automatically by `for..of`.

But we can also do it "manually":

Example

```
let iterator = myNumbers[Symbol.iterator]();

while (true) {
  const result = iterator.next();
  if (result.done) break;
  // Any Code Here
}
```

Try it Yourself »

Note

Iterators provide a controlled way to work with **data sequences**, enabling **custom iteration logic** for various data structures.

In the next chapter you will learn about the new **Iterator Helper Functions** introduced in ECMAScript 2025.

Exercise ?

```
  i++;
}
```

What will be the value of `i`?

- 0
- 1
- 2
- 3

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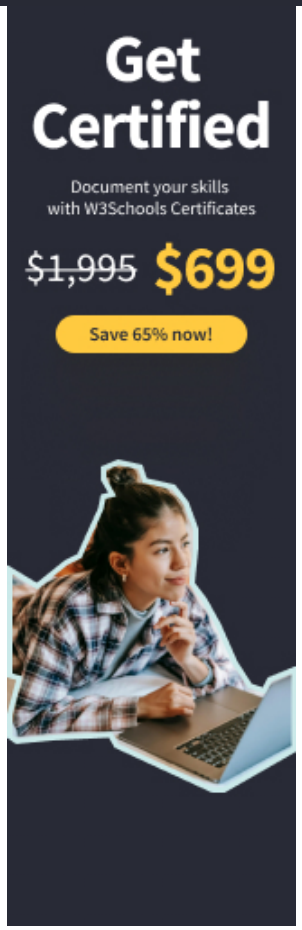

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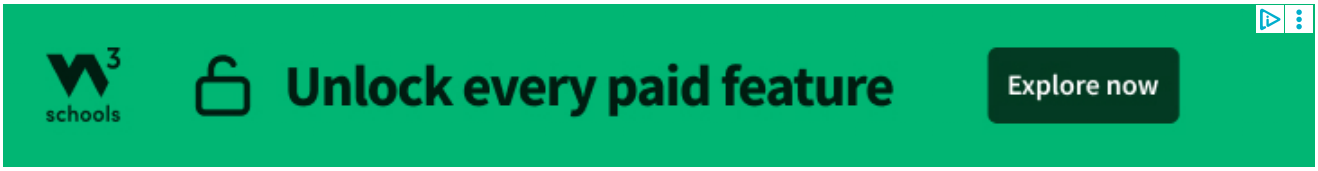

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